MTH 5000, Lecture 5

Applied Forecasting of Financial Data
Topic: Time Series Forecasting Frameworks

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What Is Forecasting?

Forecasting is a method of predicting a future event or condition by analyzing patterns and uncovering trends in previous and current data.

We do this by :

- Look at the pattern in historical data
- Use it to guess what happens next

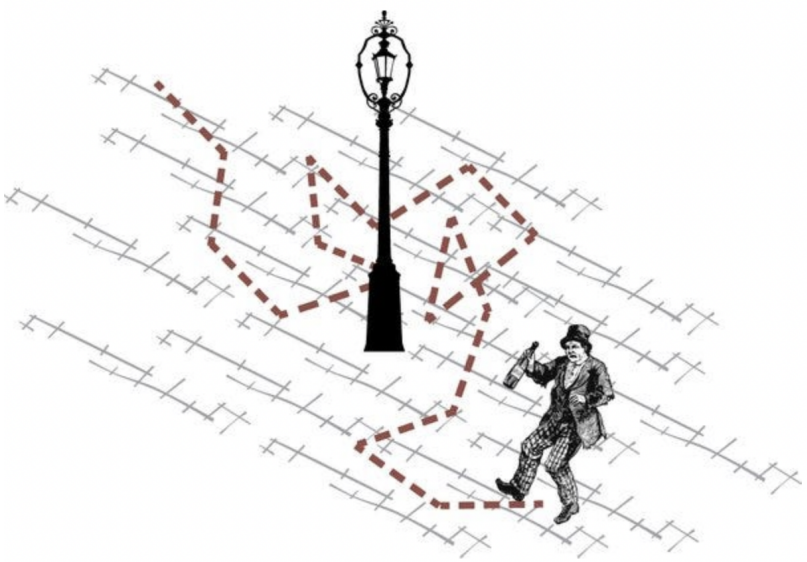
Examples :

- "How many shares of Apple will trade tomorrow?"
- "What will the closing price of Tesla be next week?"

Basically we use past patterns help inform the future, but not everything is predictable.

Forecasting: The Drunk Man Analogy

- Imagine a drunk man walking along a street.
- **Goal:** Predict where he will be in the next step.
- Observation: His previous steps provide information.
- Key ideas:
  - If he usually moves slowly forward $\rightarrow$ future is somewhat predictable.
  - If he sways randomly $\rightarrow$ future is uncertain.
- Multi-step forecasting: uncertainty grows as we look further ahead.



Forecasting: Mathematical Formulation

- Let y_t be the observed value at time t.
- Goal: Predict y_{t+h} using all available information up to time t.
- Formal definition:

$$\hat{y}_{t+h|t} = \mathbb{E}[y_{t+h} \mid \mathcal{F}_t]$$

- Where:
  - $\hat{y}_{t+h|t}$: forecast h steps ahead
  - $\mathcal{F}_t$: all information available at time t (past observations, features)
  - h: forecast horizon (1-step, multi-step)
- Intuition: *This is our "best guess" given what we know so far.*

Time Series in Relation to: Intuition

As we know Forecasting is predicting future values based on past observations.
A time series is a sequence of data points recorded at specific, usually consistent, time intervals.

- Connecting **time series** to **forecasting**?
  - In time series observations are ordered in time $\rightarrow$ past informs the future.
  - Patterns like trends, cycles, and seasonality guide predictions.
  - Noise/random variation limits perfect prediction.
- Examples:
  - Daily stock prices $\rightarrow$ forecast tomorrow's closing price.
  - Monthly sales figures $\rightarrow$ forecast next month's revenue.
  - Hourly temperature readings $\rightarrow$ forecast tomorrow's high.

Without a time series structure, forecasting reduces to guessing the average value.

Why Forecasting Requires a Time Series: Mathematical Formulation

If we let y_t be the observed value at time t our data is of the form .

- Time series: $\{y_1, y_2, \ldots, y_T\}$ which is observed at regular intervals of time.

Thus our future forecasted value:

$$y_{t+1} = f(y_t, y_{t-1}, \ldots) + \varepsilon_{t+1}, \text{ where } \varepsilon_{t+1} = \text{random noise}$$

Therefore:

- Past observations $\{y_1, y_2, \ldots, y_t\}$ inform the future.
- This sequential dependence defines a time series.
- Forecasting without temporal dependence is impossible.
- Random noise limits perfect prediction.

Time Series Decomposition

Time series decomposition is a statistical method that breaks down a time-indexed dataset into three distinct, manageable components

- **Trend-Cycle (T_t):** The long-term, persistent movement of the data, representing upward or downward shifts over a long period.
- **Seasonality (S_t):** Repeating, fixed-period, and predictable patterns (e.g., higher sales every December, weekly traffic dips).
- **Remainder / Noise (R_t):** Random, irregular, or unpredictable variations not explained by trend or seasonality.

How this improves forecasting:

- We can model each component separately.
- Trend + seasonality captures the predictable part.
- Noise represents uncertainty, which we accept but cannot fully predict.

Additive vs Multiplicative Decomposition

Instead of forecasting one messy series, we forecast our decomposed structured parts and account for randomness.

Additive Decomposition

- Formula: $y_t = T_t + S_t + R_t$
- Definition: The observed series is the sum of its components.
- Seasonal effect is **roughly constant** regardless of the trend.
- Useful when variations do not increase with the level of the series.

Multiplicative Decomposition

- Formula: $y_t = T_t \times S_t \times R_t$
- Definition: The observed series is the product of its components.
- Seasonal effect **changes proportionally** with the trend.
- Useful when variations grow as the level of the series increases.

Stock Prices: Additive vs Multiplicative Decomposition

Key Idea: Stock prices often exhibit proportional variation.

- **Additive decomposition:** $y_t = T_t + S_t + R_t$
  - Assumes seasonal/short-term effects are **constant in magnitude**.
  - Rarely true for stocks — higher prices usually have bigger fluctuations.
- **Multiplicative decomposition:** $y_t = T_t \times S_t \times R_t$
  - Seasonal and short-term effects grow proportionally with the trend.
  - Captures the fact that a \$100 stock may swing ± 5, while a \$10 stock swings ± 0.5.

Rule of Thumb: For stock prices, use **multiplicative decomposition** in most cases.